CS23302	DATA STRUCTURES	L	T	P	C
		3	0	4	5

UNIT – I LINEAR DATA STRUCTURES **10L, 12P**

Introduction to Data Structures & Algorithms – Asymptotic notations - ADT – Array - List - Linked List - Singly Linked List - Doubly Linked List – Circular List – Elementary Operations - Stack – Queue - Array Implementation – Linked List Implementation – Applications.

PRACTICALS:

1. Array implementation of Stack, Queue ADTs
2. Linked list implementation of Stack and Queue ADTs
3. Applications of Stack and Queue ADTs.

UNIT – II NON-LINEAR DATA STRUCTURES **9L, 12P**

Tree - Binary Tree – Properties – Representation - Binary Search Tree – Operation - Traversal – Threaded Binary Tree – Heap – Priority Queue implementation with heap - Graph - Terminologies – Representation – BFS - DFS - Connected Components – Minimum Spanning Tree.

PRACTICALS:

1. Implementation of Binary Trees, Traversal operations
2. Implementing Heap structure
3. Implementing search algorithms on graph

UNIT – III SORTING **8L, 12P**

Insertion Sort – Heap Sort - Merge Sort – Quick Sort – Counting Sort – External Sorting – Multi-way Merge Sort.

PRACTICALS:

1. Implementation of Sorting Algorithms

UNIT – IV SEARCHING **6L, 12P**

Linear Search – Binary Search - Hash Function – Separate Chaining - Linear Probing – Quadratic Probing – Double Hashing – Rehashing.

PRACTICALS:

1. Implementation of Searching Algorithm
2. Operations on Binary Search Trees

UNIT – V SEARCH STRUCTURES **12L, 12P**

Balanced Tree - AVL Tree – Red Black Tree – Multi-way Search Tree - B-Tree – Binary Trie – Multi-way Trie – Suffix tree.

PRACTICALS:

1. Implementing balanced BST

TOTAL: 45L + 60P = 105 PERIODS

COURSE OUTCOMES:

Upon completion of the course, the students will be able to

1. Apply efficient data structures required for an application
2. Compare sorting algorithms on time complexity
3. Understand, design and implement linear and non-linear data structures

4. Appreciate basic and optimal search structures
5. Select suitable search algorithm for an application

REFERENCES:

1. Ellis Horowitz and Sartaj Sahni, Anderson Freed “Fundamentals of Data Structures in C”, University Press, 2008.
2. Ellis Horowitz and Sartaj Sahni, Dinesh Mehta “Fundamentals of Data Structures in C++”, Silicon Press, 2007.
3. Yashavant Kanetkar, “Data Structures Through C”, BPB press, 4th edition, 2022.
4. Michael T. Goodrich, Roberto Tamassia “Data Structures and Algorithms in Python”, Wiley, 2021.
5. Jean-Paul Tremblay and Paul G Sorenson, “An Introduction to Data Structures with Applications”, Second Edition, McGrawHill, 2017.
6. Thomas H Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, “Introduction to Algorithms”, Third Edition, Prentice Hall, 2010.
7. Ellis Horowitz and Sartaj Sahni, “Fundamental of Computer Algorithms”, Galgotia, 1985.

Mapping of CO with PO

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	2	2	-	-	-	3	2	-	3	2	3	2
CO2	3	3	2	2	-	-	-	-	-	-	-	-	3	2	2
CO3	3	3	3	3	2	-	-	-	3	-	-	3	3	3	3
CO4	3	1	2	2	-	-	-	-	-	-	-	3	1	3	2
CO5	2	1	2	3	-	-	-	-	-	2	-	-	1	2	2

1 - low, 2 - medium, 3 - high, '-' - no correlation